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Climate change impacts on crop yields, land use and environment in response to crop sowing dates and thermal time requirements

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ABSTRACT

Impacts of climate change on European agricultural production, land use and the environment depend on its impact on crop yields. However, many impact studies assume that crop management remains unchanged in future scenarios, while farmers may adapt their sowing dates and cultivar thermal time requirements to minimize yield losses or realize yield gains. The main objective of this study was to investigate the sensitivity of climate change impacts on European crop yields, land use, production and environmental variables to adaptations in crops sowing dates and varieties' thermal time requirements. A crop, economic and environmental model were coupled in an integrated assessment modelling approach for six important crops, for 27 countries of the European Union (EU27) to assess results of three SRES climate change scenarios to 2050. Crop yields under climate change were simulated considering three different management cases; (i) no change in crop management from baseline conditions (NoAd), (ii) adaptation of sowing date and thermal time requirements to give highest yields to 2050 (Opt) and (iii) a more conservative adaptation of sowing date and thermal time requirements (Act). Averaged across EU27, relative changes in water-limited crop yields due to climate change and increased CO_2 varied between − 6 and + 21% considering NoAd management, whereas impacts with Opt management varied between + 12 and + 53%, and those under Act management between − 2 and + 27%. However, relative yield increases under climate change increased to + 17 and + 51% when technology progress was also considered. Importantly, the sensitivity to crop management assumptions of land use, production and environmental impacts were less pronounced than for crop yields due to the influence of corresponding market, farm resource and land allocation adjustments along the model chain acting via economic optimization of yields. We conclude that assumptions about crop sowing dates and thermal time requirements affect impact variables but to a different extent and generally decreasing for variables affected by economic drivers.

1. Introduction

Understanding how climate change may affect European arable agriculture is important to guide decision making around possible adaptations in agricultural management, to minimize damages and realize benefits (Howden et al., 2007; Ewert, 2012; Hall et al., 2012; Vermeulen et al., 2012; Kahiluoto et al., 2014). Consideration of climate change impacts on crop productivity alone is, however, not sufficient to project how cropping patterns will respond to climate change as changes in technology, prices and trade likewise all affect agricultural management (Reilly et al., 2003). While there is an increasing number of crop modelling impact studies quantifying the impacts of climatic factors on European crop productivity (Asseng et al., 2015; Webber et al., 2015; Zhao et al., 2015b), fewer examples examine how these may translate into changes in land use and production, and the resulting impact on the environment (Reilly et al., 2003; Nelson et al., 2013; Shrestha et al., 2013). Ideally, such analysis would make use of knowledge from a suite of disciplinary models, able to simulate crop productivity and farm resource allocation as well as resource use and degradation (Antle et al., 2004; van Ittersum et al., 2008; Ewert

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et al., 2009; Britz et al., 2012) appropriate for the scale, region and particular problem considered (Ewert et al., 2015).

While the response of crop productivity to climate change alone is insufficient, accurate estimates of how crop yields will respond to higher temperatures and changed precipitation patterns are critical to understanding the broader implications of climate change for agriculture. In their integrated assessment (IA) of climate change effects on poverty around the world, Hertel et al. (2010) demonstrated that future global crop commodity prices and poverty outcomes in 2030 were extremely sensitive to the magnitude of relative climate change impacts on crop yields. Likewise, in a global integrated assessment of climate change on global food systems, Nelson et al. (2013) and von Lampe et al. (2014) showed that their economic models were very responsive to the size of the climate change impact on crop yields. Collectively, these studies demonstrate the need for robust estimates of how crop yields are likely to respond to climate change.

The crop sowing date and choice of variety determine how much radiation is captured by the crop, a key determinant of potential yield levels (Van Ittersum and Rabbinge, 1997). As temperatures warm, phenological development accelerates and crop yields of a given variety can be expected to decline due to a shorter period to intercept radiation (Craufurd and Wheeler, 2009). The acceleration of phenology with warming has been observed across Europe (Menzel et al., 2006) with the result that longer growing seasons are possible for potential growth. Siebert and Ewert (2012) and Estrella et al. (2009) report accelerated development for a number of annual crops in Germany, though less pronounced for winter sown crops due to the buffering effect of vernalization and photoperiod responses (McMaster et al., 2008). Changes in crop phenology can be only partially explained by changes in climate; changes in crop management also change the occurrence of key phenological stages (Craufurd and Wheeler, 2009). For example, between 1959 and 2009, roughly one third of the advancement of phenological stages of oats in Germany can be attributed to the use of shorter season varieties, advised by extension services as a means to avoid later summer drought (Siebert and Ewert, 2012). Likewise, trends of earlier crop sowing dates with warming temperatures have also been observed (Estrella et al., 2009). Rezaei et al. (2015) report the sowing date for winter wheat in Germany advanced by 5 days in the period 1951–2009, while a subsequent study reported the sowing date for winter rye advanced by 1.3 days decade^{-1} whereas the sowing date for winter rapeseed remained largely the same (Eyshi Rezaei et al., 2017). These studies demonstrate both the influence of climate on sowing dates, crop development, and varietal choices, but also that other factors in addition to warming (e.g. risk of drought) influence farmers' varietal choices, which appears to vary with crop and region (Eyshi Rezaei et al., 2017). Few impact studies to date (for recent exceptions see, Webber et al., 2015; Zhao et al., 2015b) have quantified the uncertainty due to impacts of changes in crop management on crop yields (Müller and Robertson, 2014; Ewert et al., 2015). However, as sowing dates and variety choice are likely to vary with region and crop, as well as economic factors related to timing of operations (Eyshi Rezaei et al., 2017), their specification in climate change impact studies is not expected to be straightforward. Rather, it is likely to require iteration to select highest yield levels, or explicit assumptions about the criteria used by farmers to select desired management (i.e. highest yields versus drought risk management). In addition to affecting crop yields, crop management specification is expected to affect land use, total production and nutrient inputs to achieve these yields and thereby the losses of nutrients to the environment with potential adverse impacts (Reilly et al., 2003).

Considering both climate change impacts and the effects of specific crop management changes at all levels has rarely been done in IA assessments (an exception are Reilly et al., 2003 who assess the effects of adaptations such as irrigation water supply, pesticide use and international trade on the agricultural economy, regional crop and livestock production, irrigation water use and irrigated area, and cropland and land use). There are various literature reviews considering general adaptations in IA assessments as provided by Patt et al. (2010), Fisher-Vanden et al. (2013) and Hertel and Lobell (2014), but the "biases that are introduced when these effects are not considered in the analysis" (Fisher-Vanden et al., 2013), have hardly ever been assessed.

In this context, the overarching aim of the study was to quantify the impacts on key agronomic, economic and environmental variables for European cropping systems arising from different assumptions about crop sowing date and varietal choice (thermal time requirements) in a climate change impact study. The study was conducted with an integrated assessment modelling (IAM) framework for 27 countries of the European Union (EU27) using three disciplinary models (crop, economic and environmental).

2. Methods

2.1. Integrated modelling approach

A suite of three disciplinary models, the crop-growth model in the SIMPLACE framework (Gaiser et al., 2013) combining the models Lintul-5 (Wolf, 2012), DRUNIR (Spitters and Schapendonk, 1990; Van Oijen and Lefelaar, 2008) and Heat (Eyshi Rezaei et al., 2013), the economic agricultural sector model CAPRI (Britz et al., 2006; Britz and Witzke, 2014) and the environmental impact model INTEGRATOR (de Vries et al., 2011; Kros et al., 2012), were applied together for Europe in an integrated assessment modelling (IAM) exercise. Individual models conducted simulations for a baseline period, as well as, under different climate change and socio-economic scenarios centered around 2050. While the spatial and temporal resolution, the extent of input data and the base simulation unit differed between models, results were aggregated to and passed between models at the level of the NUTS2 administrative regions (see http://epp.eurostat.ec.europa.eu/portal/page/portal/nuts_nomenclature/introduction). The main input and output relations between the data sources and models are shown in Fig. 1. SIMPLACE simulated water-limited growth and development of six crops in response to climate, CO_2 concentration, and crop management (sowing dates and varietal thermal time requirements) in three specifications. Relative yield changes due to climate change and management adaptation were determined for each scenario, relative to the SIMPLACE baseline, and passed into the economic agricultural sector model CAPRI. CAPRI also requires an input on the relative yield changes due to technology progress, and this was estimated using historical yield trends, modified for each SRES scenario following the approach of Ewert et al. (2005). The two sources of yield change were summed and passed as input to CAPRI as exogenous yield shocks.

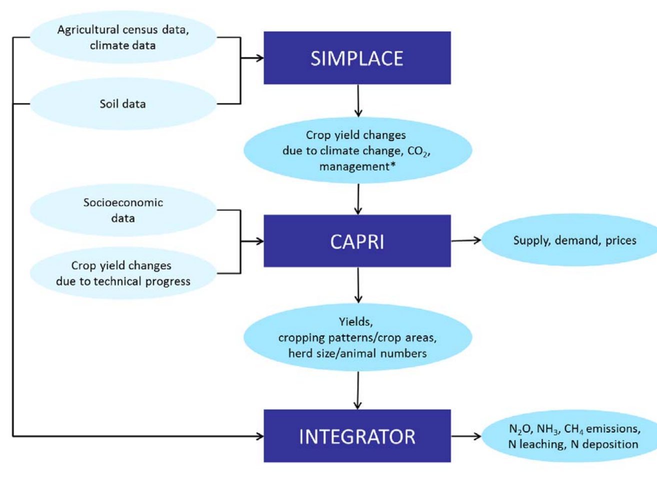

Fig. 1. Information flow between data sources and models including simulated impacts as used and simulated in this study.

Likewise, SRES scenario specific changes in global gross domestic product (GDP), population, and agricultural trade policies were considered, to simulate changes in European agricultural markets (supply, demand and prices). Based on these and the other scenario information, yields were endogenously readapted in CAPRI. The resulting changes in crop yield, herd sizes, and cropping areas simulated by CAPRI were used as input to INTEGRATOR to derive changes in fertilizer and manure N input and agricultural N losses to air and water (Fig. 1). Detailed descriptions of SIMPLACE, CAPRI and INTEGRATOR are given in the supplementary material.

2.2. Scenarios and reference run

Assumptions about climate change and socioeconomic conditions were based on the Special Report on Emissions Scenarios (SRES) (Nakicenovic and Swart, 2000).[2] The SRES scenarios link global atmospheric CO_2 concentration to demographic, societal, economic and technological storylines, for which quantitative macroeconomic global scenarios are specified (population and GDP growth by region). The CO_2 concentration assumed in each storyline corresponds to that assumed by the SRES scenarios used to drive global circulation models (GCMs) which simulate future climatic variables (temperature, precipitation). The SRES scenarios considered in this study are A1B, B1 and B2. Comparisons of the SRES scenarios to the newer representative concentration pathways (RCPs) and shared socioeconomic pathways (SSP) relevant to this study are provided by van Vuuren and Carter (2014).[3] The GCM outputs, together with the corresponding CO_2 concentrations and related demographic and key economic changes were used as input to the various components of the IAM. Global yield changes needed for CAPRI for the A1B, B1 and B2 scenarios were taken from Parry et al. (2004) to complement the SIMPLACE yield changes for Europe. A summary of the scenarios considered, with key assumptions and data, is presented in Table 1 following Wolf et al. (2015).

In addition to the changes in exogenous yields, population, GDP and trade policy, changes were considered in the economic assessment. Compared to A1B and B2, B1 is an intermediate scenario with medium global population growth and medium world GDP growth until 2050. Current agricultural and trade policies were assumed to continue. A1B also features medium global population growth by 2050. However, world GDP was assumed to be much higher as in the B1 and B2 scenarios and a trade liberalization including a reduction of tariffs and an expansion of Tariff Rate Quotas (TRQs)[4] was applied. B2 assumes higher population growth and lower global income as in the B1 scenario. Whereas the scenario results of SIMPLACE and INTEGRATOR were compared to a base year, i.e. a year in the past (the years 2004 and 2008), the CAPRI results were compared to a reference run[5] (here for the year 2050). In terms of scenario assumptions, B1 is closest to the CAPRI reference run. It differs from the reference run by considering climate change.

[2] The SRES scenarios instead of the newer representative concentration pathways (RCPs) and shared socio-economic pathways (SSPs) of the 5th IPCC Assessment Report are used as this study was part of a larger European project which started before the RCPs and SSPs were developed.

[3] Van Vuuren and Carter (2014) provide a comparison between the SRES and the RCPs and SSPs. Although the relationships between the different sets of scenarios are complex, they conclude that A1B is located between RCP6.0 and RCP8.5 and roughly similar to SSP2, B1 is roughly similar to RCP4.5 and SSP1, and B2 is positioned in-between RCP 4.5 and RCP 6.0 and SSP2.

[4] During WTO negotiations (2009) Ambassador Crawford Falconer, chairperson of the agriculture negotiations, circulated his latest revised draft "modalities" text. This text contained formulas for cutting tariffs and trade-distorting subsidies and related provisions. For more information, see: https://www.wto.org/english/tratop_e/agric_e/chair_texts08_e.htm.

[5] In the economic literature the reference run is usually referred to as baseline or baseline scenario. We try to avoid this term in the paper to prevent confusion, as the baseline in the natural science models is conceptually different in that it describes a time horizon in the past.

2.3. Technology trends in regional crop yields

Trends in crop yields due to changes in technology up to 2050 were calculated as a continuation of historical yield levels as reported at NUTS2 level. Depending on the SRES scenarios, different assumptions were made about the propagation of the historical yield trends, following the approach in Ewert et al. (2005), as used in Hermans et al. (2010). Inherent in the approach is the division of future yield trends into two parts: progress in breeding (yield potential) and progress in management (yield gap), which are allowed to evolve differently according to the SRES scenarios. As the variability of observed yields generally increases at finer spatial resolution with resulting non-significant slopes at sub-national scales, a procedure was developed to ensure robust trends by combining trends at different levels of spatial aggregation based on their respective coefficients of determination (R^2). Trends and the corresponding coefficient of determination were calculated at the level of NUTS2, country and EU 15 grouping (i.e. one of either: an original member of the former EU 15; or a current EU member that joined the EU after the original 15 member states). If the slope was significant ($p = 0.05$) at the NUTS2 level, it was used as the historical yield trend. In the case the slope was not significant, the trend, b, for each NUTS2 zone was estimated by considering also trends at the national (NUTS0) and EU 15 grouping levels, weighted by their respective R^2 values, as:

$$b = R_{NS2}^2 \frac{R_{NS2}^2}{R_{NS2}^2 + R_{NS0}^2} b_{NS2} + \left(1 - R_{NS2}^2 \frac{R_{N2}^2}{R_{NS2}^2 + R_{N0}^2}\right)\left[R_{N0}^2 \frac{R_{N0}^2}{R_{N0}^2 + R_{EU}^2} b_{N0} + \left(\left(1 - R_{N0}^2 \frac{R_{N0}^2}{R_{N0}^2 + R_{EU}^2}\right) b_{EU}\right)\right]$$

where R^2 is respective the coefficient of determination for a linear regression of historical yields per crop, the indices for NS2, NS0 and EU referring to yield aggregates at the NUTS2, NUTS0 (i.e. national) and EU15 level, respectively.

2.4. Sowing date and variety specification

Three management situations about crop varietal choice and sowing dates were evaluated with SIMPLACE for the baseline period and for each future scenario (A1B, B1 and B2). Sowing dates were adjusted by (− 30 days, − 20 day, − 10 days, no change, + 10 days) relative to the current practice as reported for each crop and environmental zone in the JRC Mars database (https://ec.europa.eu/jrc/en/mars). Variety was adjusted by consideration of thermal time requirements for development to maturity (+ 30%, + 20%, + 10, no change, − 10%) relative to the values obtained by simulating crop development from emergence to maturity under baseline climate conditions. Baseline conditions were based on observed sowing, anthesis and harvest dates for each crop in each environmental zone in the JRC Mars database.

The three management situations for each future scenario period were evaluated, using the following designations: no adaptation (NoAd) of sowing dates and variety; simulated optimized adaptation (Opt) of sowing dates and variety; and an intermediate case, termed actual adaptation (Act). For NoAd simulations, sowing dates and thermal time requirement remained unchanged between the future and baseline periods for each simulation unit. For the Opt case, the selected management was based on the combination of sowing date and thermal time requirement that gave the highest yield in the future period. For simulations in the baseline period, management was specified as the current practice. The Opt case results from a simulation optimization process may differ considerably from famers' optimal management. For the Act simulation case, the relative yield change Opt was reduced by the relative yield increases in the baseline period that result when baseline sowing dates and thermal time requirements are selected to give the highest yield (in the baseline) compared to the baseline yields

Table 1
Description of the scenarios considered, key assumptions and related data for 2050.

	A1B	B1	B2
Exogenous assumptions	Inflation rate of 1.9% per year Constant exchange rates World GDP: 181 Trillion 1990 US$/year World population: 8.7 billion	World GDP: 136 Trillion 1990 US$/year World population: 8.7 billion	World GDP: 110 Trillion 1990 US $/year World population: 9.3
Commodity prices	CAPRI simulation results		
Input prices	Oil price estimated as a function of GDP growth, input costs of agricultural activities updated according to energy cost share		
Yield global	Parry et al. (2004) SRES A1B	Parry et al. (2004) SRES B1	Parry et al. (2004) SRES B2
Yield EU (with management specifications)	SIMPLACE simulation (15-GCM ensemble mean)	SIMPLACE simulation (MIROC3.2 GCM)	SIMPLACE simulation(15-GCM ensemble mean)
Set-aside and quota policies	Abolishing obligatory set-aside, expiry of mild and sugar quotas		
Premium scheme	2009 health check (decoupled payments, increased modulation)		
WTO trade policy	Reduction of tariffs and expansion of TRQ (sensitive products) as proposed by Falconer (2009)	Tariffs and TRQ as in 2008	

resulting from current crop management. This optimization of sowing date and thermal time requirements in the baseline period provided an estimate of the amount by which management in the baseline period for each zone as currently practiced deviates (on average) from the optimized value as simulated by SIMPLACE. Deviation is expected as (1) default sowing dates and crop cultivars were defined for 13 environmental zones, whereas the optimization was conducted for each simulation unit, (2) actual farming practices are constrained by bio-physical factors not considered in the current solution of SIMPLACE (e.g. soil workability in spring) and (3) actual farming practices are additionally determined by economic and cultural factors not considered in crop models (e.g. competing labor demands).

SIMPLACE evaluated the effects of NoAd, Opt and Act crop management under climate change on relative crop yield changes between the scenario period and the baseline period. CAPRI evaluated the effects of Opt versus Act on deviations of prices, production, land use and consumption from the reference run. INTEGRATOR evaluated the effects on nitrogen inputs from fertilizers and manure, and nitrogen losses to the environment between the scenario period and the baseline. In CAPRI and INTEGRATOR only the Opt and Act effects were considered as the NoAd case, in which farmers in 2050 would maintain the sowing dates and varieties from 2004, is considered very unlikely in the economic analysis. Furthermore, pure climate change and economic scenario impacts have been investigated in other studies (e.g. Nelson et al., 2013 and von Lampe et al., 2014).

3. Results

3.1. Changes in crop yields

3.1.1. Statistical yield trends

In accordance with other studies, annual yield increases of most crops in Europe were smaller than those during the past decades (Finger, 2010; Ray et al., 2012). The EU average relative annual yield increases for six crops simulated by SIMPLACE and the three scenarios are presented in Table 2. In most cases, relative yield increases were less than 0.8% per year, with the largest relative increases expected for the A1B and B1 scenarios in accordance with their higher overall economic growth. There was relatively little variation in the distribution of yield trends across Europe (results not shown).

3.1.2. Sowing date and variety selection for optimized crop yields

The crop management resulting in the highest yields for SRES scenario A1B in the period centered on 2050 are shown for all crops in Fig. S1 in the supplementary material while the results for maize and winter wheat are presented in Fig. 2. In general, sowing is earlier relative to current dates across Europe whereas the selection of thermal time requirements in new varieties varies spatially with prevailing climatic condition (Fig. 2). For maize, this implies shorter season varieties in parts of Southern Europe (Spain and Italy) and longer season varieties in large parts of Northern, Central and Eastern Europe. The selection of shorter season varieties for maize in the South may be overstated here as they largely result due to increased heat stress. Maize is largely irrigated in these regions and our model neglects the cooling effects of irrigation (Webber et al., 2016b) shown as critical in studies of irrigated maize in Spain (Gabaldón-Leal et al., 2016). Of course, our simulations have assumed the intensity of irrigated production is constant. For winter wheat, which is largely rainfed, the simulations indicate that the highest yields are achieved when shorter season varieties are adopted, which may indicate either or both drought or heat stress increase later in the summer. However, optimization of current sowing dates and thermal times (Fig. 3), suggest that yields would increase under current climatic conditions if winter wheat thermal requirements were shorter and sowing earlier for large parts of Europe. For maize, on the other hand, current thermal time requirements seem largely adequate, though simulated yields would improve with earlier sowing.

Table 2
Relative EU27 yield increases (%) for six crops between 2004 and 2050 due to technology progress as estimated by the propagation of historical yield trends for three SRES scenarios modified according to assumptions about breeding (yield potential) and improved crop management (closing the yield gap) advances.

Crop	Scenario		
	A1B	B1	B2
Wheat	35	28	19
Barley	37	30	20
Grain maize	50	40	27
Potato	32	25	17
Rapeseed	33	27	18
Sugar beet	51	40	27

Impact of climate change and crop management assumptions on crop yields.

Relative changes in EU27 averaged simulated water-limited yields due to climate change and increased CO_2 with NoAd varied between − 6% to + 21% (Table 3), considerably less than the potential yield increases due to technological progress as indicated by the annual yield trend in Table 2. Averaged across Europe, yields of grain maize and potato change varied between − 4 to 3% and − 6 to 3%, respectively, with the most optimistic yield changes in the B1 scenario and the most negative yield changes in the B2 scenario (Table 3). For the winter sown cereals (wheat and barley), rapeseed and sugar beet, climate change resulted in yield increases between + 4% and + 21%, with the largest increases associated with the A1B scenario. These yield increases were largely due to the positive impact of increased CO_2 which effect is greater for the C3 crops (all crops except maize) than for C4 crops (grain maize). As with grain maize and potato, the least positive yield

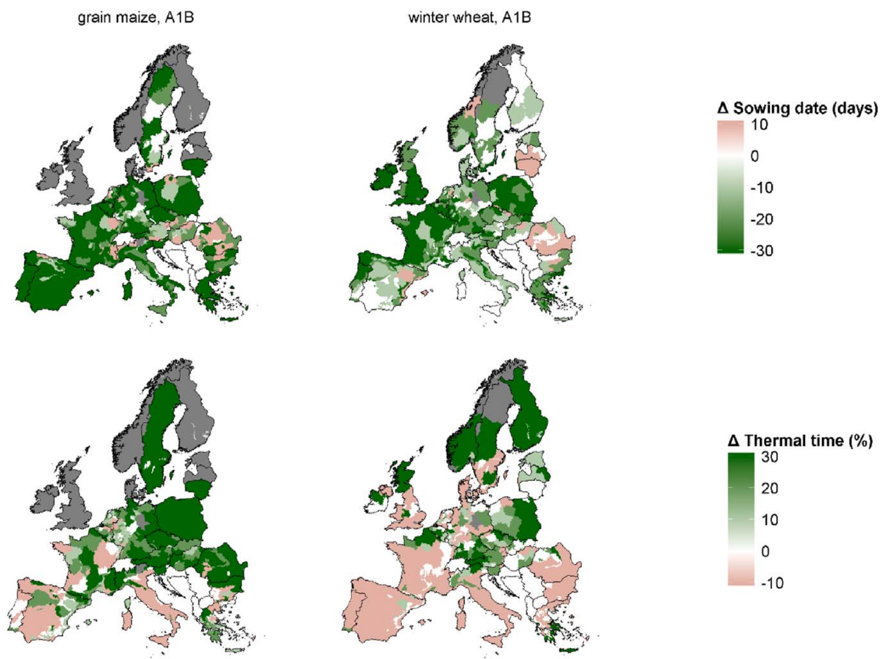

Fig. 2. Adaptations in sowing date (+ indicating days earlier, 0 for unchanged, and - indicating days later) (top row) and thermal requirements (% relative to the thermal time requirement in the baseline period) (bottom row) for grain maize (left hand column) and winter wheat (right hand column) for the A1B scenario for the period 2040 to 2064.

changes for these crops occurred for the B2 scenario. However, the distribution of climate effects was not uniform across Europe; for Southern Europe we simulated very large yield decreases for potatoes and maize, with smaller but still negative yield impacts on winter cereals in Spain, as reported in Webber et al. (2015) and Zhao et al. (2015).

When sowing dates and crop cultivars were optimized (Opt) to give the highest water-limited yields under climate change conditions in 2050, the EU average yield increases were much higher for most crops and scenarios, particularly for winter sown cereals (Table 3, Fig. 2). This resulted from the combined adjustment of sowing dates and use of varieties with thermal time requirements to maximize yield under the higher temperature regime, with possibly new timing and severity of water shortages. When actual adaptation (Act) was considered, the EU average relative yield changes remained positive but less than in the case of Opt (Table 3, Act, Fig. 3); EU27 average relative yield changes for the base period when optimized sowing dates and crop cultivars are used as compared to current baseline management are given in table S4 in the supplementary material). Even with both Opt and Act adaptations, the impact of climate change on yield changes tended to be negative in the south of Europe, especially in the case of Act adaptations. The exception to this was the yield response of sugar beet in Spain with Opt adaptation, which is very large and positive.

Examination of the relationship between the Opt and Act adaptations revealed for some crops and situations (i.e. wheat, grain maize and rapeseed) that current sowing dates and thermal time requirements (as assumed in this study) were near optimal in most cases across Europe (Fig. 4) as indicated by close agreement between Opt and Act for those crops (Figs. 5 and 6).

The above results compared the relative yield changes of the

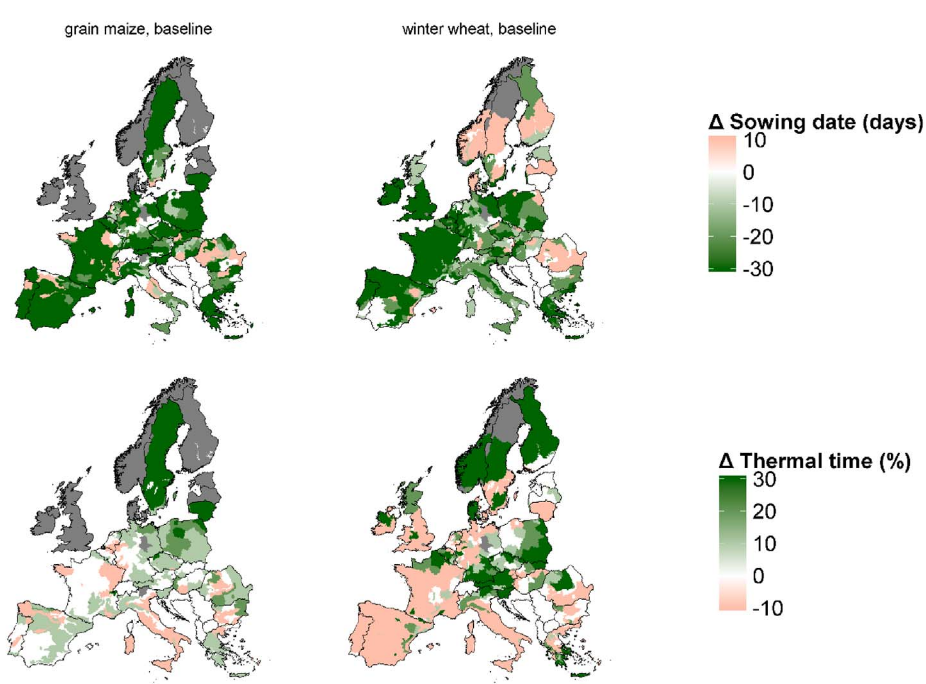

Fig. 3. Adaptations in sowing date (+ indicating days earlier, 0 for unchanged, and - indicating days later) (top row) and thermal requirements (% relative to the thermal time requirement in the baseline period) (bottom row) for grain maize (left hand column) and winter wheat (right hand column) in the current baseline climate (1982 to 2006).

Table 3
European (EU27) average water-limited relative yield changes for six crops between 2004 and 2050 due to climate change and elevated CO2, considering no adaptation (NoAd), optimized adaptation in sowing dates and crop varieties (Opt) selected to give the highest yields in the scenario period and "actual" adaptation (Act) calculated as the relative yield change for optimized adaptation minus the relative change in yields if baseline management is optimized.

Crop	Crop yield impacts in 2050 relative to 2004 (%)								
	A1B			B1			B2		
	NoAd	Opt	Act	NoAd	Opt	Act	NoAd	Opt	Act
Wheat	15	23	15	14	21	14	8	16	8
Barley	21	53	27	14	44	19	12	42	17
Grain maize	− 2	15	6	3	16	7	− 4	12	3
Potato	− 2	28	4	3	31	7	− 6	22	− 2
Rapeseed	19	34	24	13	28	18	11	25	15
Sugar beet	12	35	24	11	28	17	4	25	13

different adaptation situations considering climate change and CO_2 effects only. When technology progress was additionally considered, relative yield changes became large and positive for all crops and scenarios (Table 4). The large relative differences between the NoAd, Opt and Act yield changes with only climate change was greatly diminished when yield changes arising from both climate change and technology progress was considered (Table 4), as due to the technology progress relative yield changes arising from the continuation of historical yield trends were all positive.

3.2. Economic adjustments in farm management

As a result of the lower SIMPLACE simulated crop yields in the Act management case compared to the Opt case, endogenous yields in the economic model also tended to be lower in the Act as compared to the Opt case (Table 5). Following the only slight crop model yield increases of grain maize compared to the other crops, grain maize yields in the economic model were generally much lower than in the reference scenario.

Overall, land use changes (Table 6) were moderate in B1 compared to the reference run such that yield changes (Table 5), translated directly into changes in the production quantity (Table 7). Compared to the Opt case, the lower Act yields, though partly compensated for by increased land use, still translated into lower production quantities (Table 8).

Compared to B1, changes were more extreme in B2. Surprisingly, yields, land use and production quantities tended to decrease in the B2 scenario which features the most extreme population growth of the considered scenarios. However, the effect of the global population growth seemed to be offset by the comparably low GDP growth in this scenario. This would particularly explain low meat and cow milk production since consumption of luxury goods would decline first. As a result of the generally lower crop model Act yield changes, Act yields after economic optimization were still lower than under Opt management in the B2 scenario. As in B1, this led to higher land use for the specific crops. However, as in B1, even the increased land use did not fully compensate for the lower Act yields, such that final supply stayed lower as it would have been under Opt management.

In the A1B scenario with moderate global population growth, but considerably increasing GDP, a slight increase in total land use compared to the reference scenario was projected. Production quantities of meat and cow milk increased considerably in response to increased demand from global GDP growth. Cereal yield decreased by − 3.6% or − 4.2% for Opt and Act crop model yields, respectively, but area and production quantity increased considerably. Grain maize in the A1B scenario was the only example, where yields after economic optimization were higher in the Act than in the Opt management case.[6] As in the B1 and B2 scenarios, Act yields after economic optimization in A1B were considerably lower than Opt yields, except for grain maize. However, other than in B1 and B2, this did not necessarily translate into higher land uses under the Act management case compared to the Opt case. Following both, lower yields and lower land use under Act management in A1B, the final supply of the considered products was also lower than under Opt management. Examination of the spatial distribution of production under the Act management case in the A1B scenario revealed that changes in supply were different across regions and products (Fig. S2 in the Supplementary material).

3.3. Combined scenario impacts on nitrogen fluxes

The effect of the scenarios and management specifications on the nitrogen fluxes are presented as the relative change compared to N fluxes for the reference year (i.e. the year 2008 in this case). Results are given for both the actual (Act) and the simulated optimized (Opt) management cases. Results of average relative changes in N inputs and N outputs between 2008 and 2050 for EU27 are presented in Table 10. Results show a strong increase in N fertilizer input, being highest in the A1B scenario and lower in the direction of the B1 scenario and then the B2 scenario. This result is in line with the changes in increase in crop yields thus requiring more nutrients to produce those yields, while changes in cropping areas also play a role. Unlike N fertilizer use, N manure inputs were projected to decrease except for the A1B scenario. This was due to the estimated decrease in animal numbers for the B1 and B2 scenario in accordance with the predicted decrease in meat and milk production for those scenarios (see Table 7).

Average changes in N_2O emissions for EU27, which varied from approximately 1% for the B2 scenario to approximately 10% for the A1B scenario, mainly reflected the mean changes of the combined N fertilizer and manure inputs since the used N_2O emission factors were rather comparable for N inputs from applied fertilizers and animal manure. In contrast, changes in NH_3 emissions, which varied from approximately − 7% for the B2 scenario to approximately + 7% for the A1B scenario, mainly reflected the mean changes of the N manure inputs, since the used NH_3 emission factors were much higher for N inputs from animal manure than from applied N fertilizers. We furthermore assumed that NH_3 emission fractions decreased to 2050 due to the implementation of low emission housing systems and manure application techniques.

The impacts of different scenarios on fertilizer and manure inputs were clearly higher than the effects of the Opt versus the Act management case. Effects were nevertheless significant for the input of fertilizer, being clearly higher for the Opt than the Act management case. This was due to higher crop yield changes in the Opt case, thus requiring more nutrients to produce those yields. Impacts on the N losses due to NH_3 emission, N_2O emission and N leaching + runoff were, however, only slightly influenced by the Opt versus Act management case. The small effect as compared to the scenario used is because the latter also involved higher changes in cropping areas, apart from changes in N inputs, being apparently of more effect on the N outputs.

Maps in Fig. 7 show the geographical differences throughout the

[6] Closer examination of the grain maize results in A1B shows that, as expected, higher fertilizer application rates led to the higher yields under Act management than under Opt conditions. Likewise, farmers' income earned from grain maize yields was higher for Act conditions, which not only explains a higher cropping intensity, but also a higher cropping share of grain maize. Human consumption was slightly lower, but grain maize as feed use was considerably higher as was grain maize processing (including for biofuels) and trade (income, feed use, processing and trade not shown). These processes could indicate a relatively better economic performance of grain maize compared to the other cereals under Act conditions in A1B due to which grain maize worked as a substitute for barley, oats and other cereals, whose production was considerably reduced (not shown here).

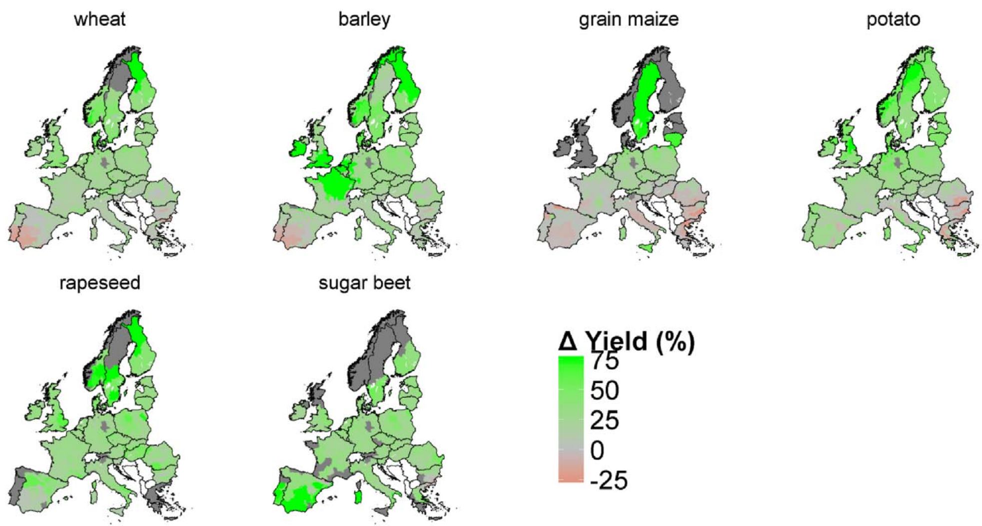

Fig. 4. Geographic variation of relative changes in simulated water-limited yields for six crops between 2004 and 2050 due to climate change and increased atmospheric CO_2 concentrations for the Opt adaptation case, i.e. optimized selection of sowing dates and thermal time requirements to give highest yields in the 2050 period for the A1B scenario.

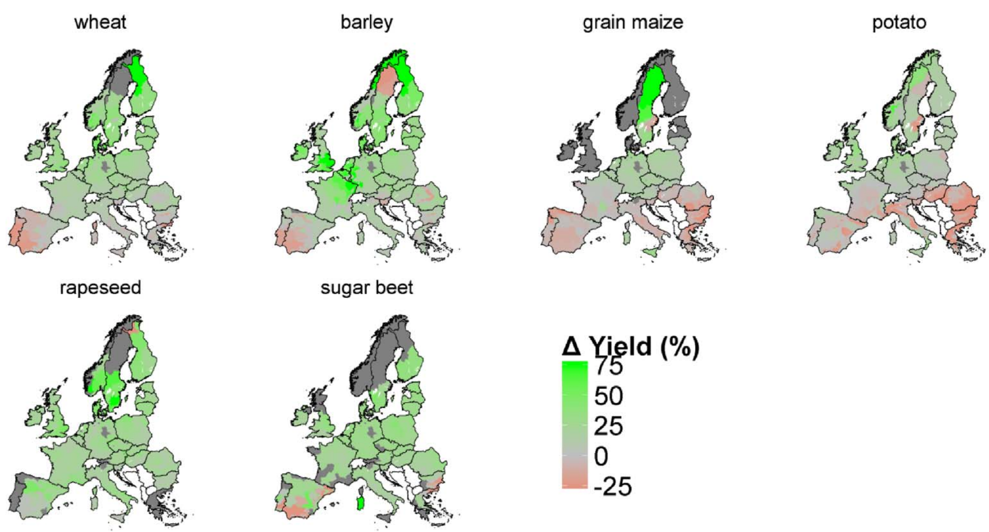

Fig. 5. Geographic variation of relative changes in simulated water-limited yields of six crops between 2004 and 2050 due to climate change and increased atmospheric CO_2 concentrations for the Act adaptation case for the A1B scenario.

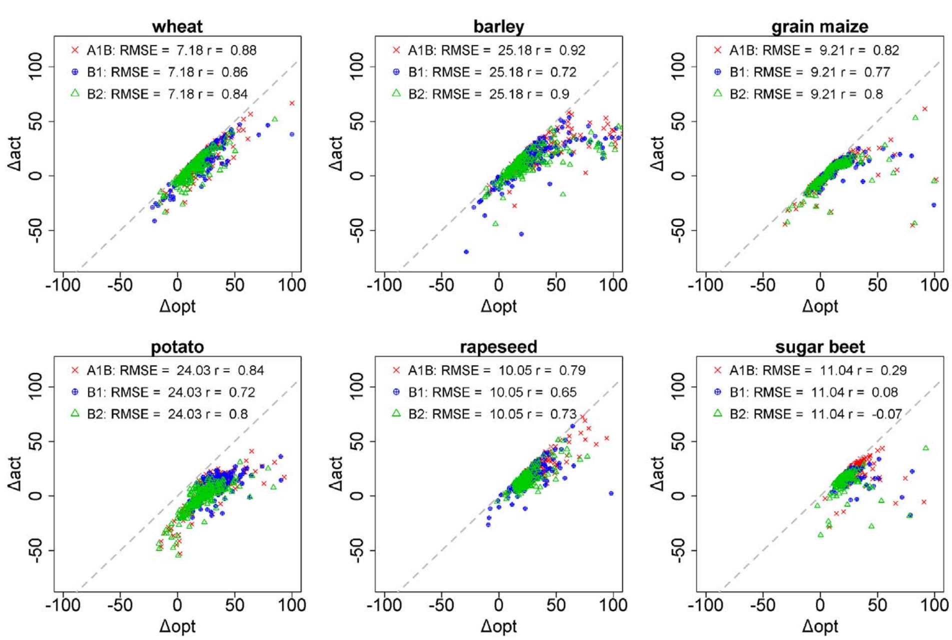

Fig. 6. Correlation between simulated relative changes in crop water-limited yield for six crops across Europe between 2004 and 2050 for optimized (Δopt) and actual (Δact) management cases for the three climate change scenarios across Europe.

Table 4
European average water-limited relative yield changes (%) for six crops across Europe between 2004 and 2050 due to climate change, elevated CO_2 and the technology trend for three SRES scenarios and three adaptation cases considering no adaptation (NoAd), optimized adaptation in sowing dates and crop varieties (Opt) selected to give the highest yields in the scenarios period and "actual" adaptation (Act) calculated as the relative yield change for the optimized adaptation minus the relative change in yields if baseline management is optimized.

Crop	Crop yield impacts in 2050 relative to 2004 (%)								
	A1B			B1			B2		
	NoAd	Opt	Act	NoAd	Opt	Act	NoAd	Opt	Act
Wheat	50	58	51	42	50	43	27	34	27
Barley	58	89	64	44	73	48	32	62	36
Grain maize	48	65	56	43	56	47	23	38	29
Potato	30	60	35	28	57	33	11	38	14
Rapeseed	52	68	58	40	55	45	29	43	33
Sugar beet	63	86	74	51	68	57	31	51	40

Table 5
CAPRI endogenous yield changes (%) of selected crops in three different SRES scenarios for the optimized (Opt) and actual (Act) management cases compared to the reference run for EU 27.

Crop	Reference	Crop yield change relative to reference (%)					
	Yield	A1B		B1		B2	
	[t/ha]	Opt	Act	Opt	Act	Opt	Act
Wheat	7	6	3	2	0.3	− 6	− 8
Barley	6	19	4	14	2	8	− 3
Grain maize	9	− 15	4	− 16	− 19	− 21	− 24
Potato	37	16	0.6	17	2	9	− 7
Rapeseed	4	− 7	0.1	− 9	− 15	− 12	− 19
Sugar beet	75	15	0.1	9	5	6	3

Table 6
Projected land use changes (%) of selected crops in the different scenarios for the optimized (Opt) and actual (Act) management cases compared to the reference run for EU 27.

Crop	Reference	Land use change relative to reference (%)					
	Land use	A1B		B1		B2	
	[million ha]	Opt	Act	Opt	Act	Opt	Act
All	177	0.9	0.9	− 0.3	0.6	− 1	− 0.2
Cereals	53	5	4	0.8	2	− 2	− 0.6
Wheat	22	5	4	1	3	− 3	− 0.9
Barley	11	6	5	0.6	2	− 3	− 0.6
Grain maize	9	7	8	0.0	2	− 4	− 3
Potato	1	3	6	− 5	− 0.3	− 10	− 5
Oilseeds	13	4	2	2	3	2	3
Rapeseed	8	4	3	2	3	2	3
Sugar beet	2	− 1	0.0	− 0.4	− 0.5	− 2	− 2

EU27. As with N_2O emissions, changes in N leaching mainly reflected the changes of the combined N fertilizer and manure inputs, as presented in Fig. S2 in the supplementary material. The relative changes in NH_3 emissions for the different scenarios and variants compared to those in the base year, showed very similar patterns to those for the N manure inputs (see Fig. S2), whereas the relative changes in N_2O emissions and N leaching plus runoff showed very similar patterns to those for the sum of N fertilizer and N manure inputs (see Fig. S2). The N fertilizer inputs in turn reflected mainly changes in crop yields and cropping patterns while N manure inputs mainly reflected changes in meat and milk production. Mostly the change varied between 25% either as an increase or decrease, with clear increases for all N losses in Sweden and France.

Table 7
Change in production (%) for selected crops in the different scenarios for the optimized (Opt) and actual (Act) management cases compared to the reference run for EU 27.

Product	Reference	Production change relative to reference (%)					
	Quantity	A1B		B1		B2	
	[million t]	Opt	Act	Opt	Act	Opt	Act
Cereals	354	8	8	0.1	− 3	− 10	− 12
Wheat	155	11	7	3	3	− 8	− 9
Barley	65	27	9	15	4	5	− 4
Grain maize	80	− 9	12	− 16	− 17	− 25	− 26
Potato	48	20	7	11	1	− 3	− 12
Oilseeds	44	− 3	3	− 7	− 13	− 12	− 18
Rapeseed	32	− 3	3	− 7	− 12	− 11	− 16
Sugar beet	147	13	0.1	8	4	4	1
Meat	46	12	12	0.0	− 1	− 5	− 6
Cow milk	206	6	5	0.2	− 0.4	− 8	− 8

Table 8
Average relative changes (%) in N inputs and N outputs for EU27 between 2008 and 2050.

N flux	Relative N flux change (%)					
	A1B		B1		B2	
	Opt	Act	Opt	Act	Opt	Act
N Inputs						
N fertilizer	22	21	17	12	10	5
N manure	5	4	− 0.9	− 2	− 6	− 8
N output						
NH_3 emission	7	7	0.6	− 2	− 5	− 7
N_2O emission	10	9	6	5	2	0.2
N leaching + runoff	9	9	5	4	0.5	− 0.8

4. Discussion

While it is commonly assumed that farmers are expected to select sowing dates and cultivar thermal time requirements as season length and average conditions gradually change with warming (Moore and Lobell, 2014), observational evidence suggests that degrees of risk avoidance, timeliness of operations, and interactions between crop management and climatic conditions will render assumptions about future sowing dates and varietal choices highly uncertain (Siebert and Ewert, 2012; Eyshi Rezaei et al., 2017). This is one of the first studies in which the sensitivity of different cross-disciplinary impact variables, such as crop yields, land use, production and environmental variables, to assumptions about future crop management have been investigated. The study has also allowed for further insights into various challenges and important drivers of change to be addressed in future IA of climate change on agriculture.

4.1. Uncertainty in the integrated assessment approach

The study contained numerous sources of uncertainty. Some of these sources of uncertainty were captured and quantified in the range of climatic and socio-economic scenarios considered in the SRES scenarios, as well as the three crop management cases considered here and discussed more fully below. Individual models may be uncertain due to structural and/or parameter uncertainty (Wallach et al., 2013) that has been investigated in many recent crop model intercomparisons (Asseng et al., 2013; Bassu et al., 2014), as well as uncertainty associated with input data (Wallach et al., 2013) and data scaling methods (Ewert et al., 2011). However, in any IA study, uncertainty is additionally introduced due to both the inability to perfectly understand and represent real, complex systems (Refsgaard et al., 2007; Gabbert et al., 2010), potentially neglecting entire processes and likely missing important

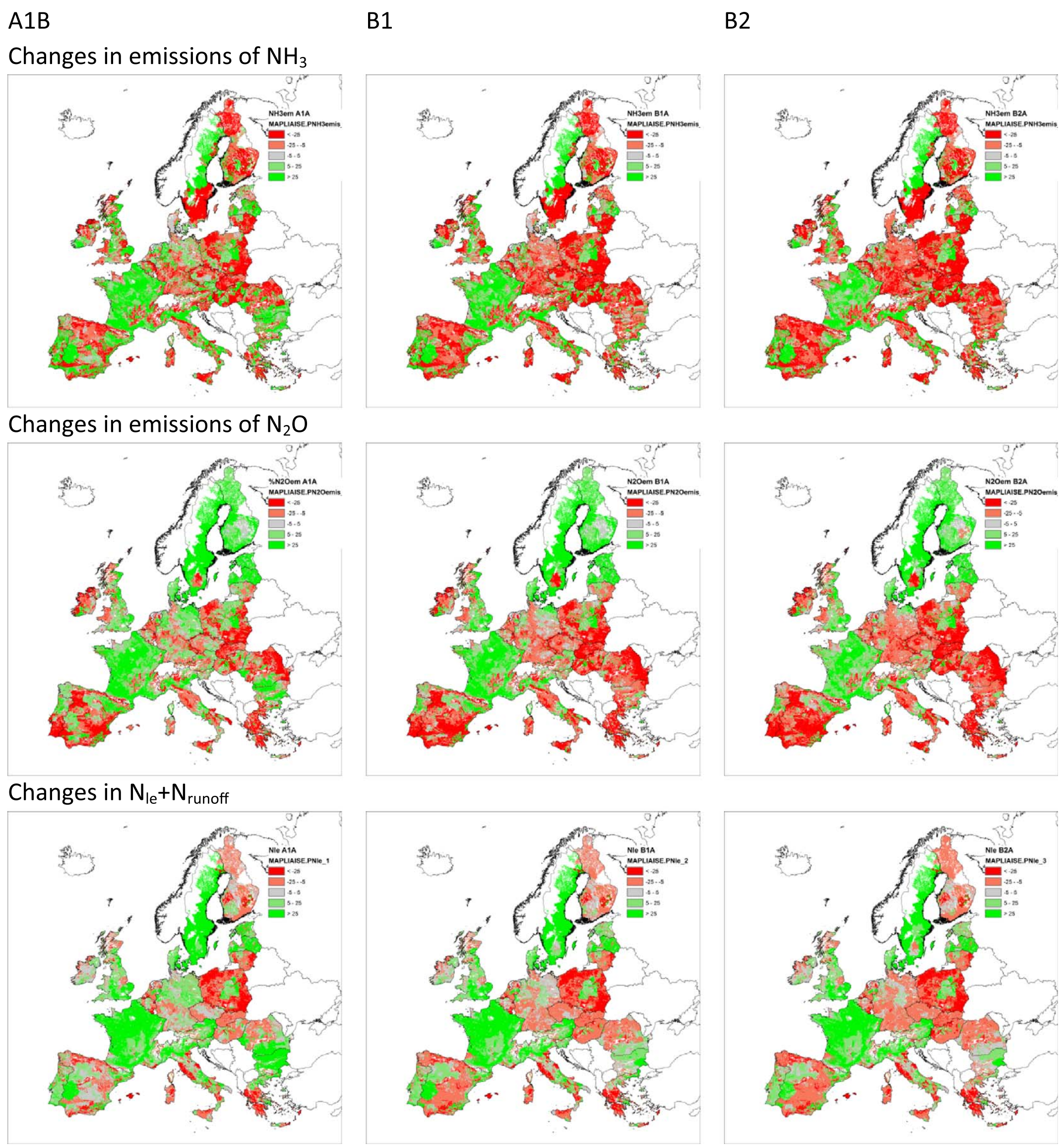

Fig. 7. Geographical distribution of the relative changes (%) in emissions of NH_3 and N_2O to air and N leaching and runoff to ground and surface water for the three scenarios with actual adaptation between 2008 and 2050.

feedbacks and interactions (Ewert et al., 2015). The current study is limited in not allowing for capturing many relevant feedbacks, ranging from interactions of nutrient losses and use by plants, to not feeding back information on the economic yield optimization to the crop yield simulations due to a combination of conceptual and methodological challenges (Janssen et al., 2011; Britz et al., 2012; Ewert et al., 2015). For example, in evaluating adaptations to climate change, crop models are used to quantify the yield benefit of the new seeds (thermal times) and management (sowing date). In our IA, the resulting yield change is subsequently used by the economic model which is additionally driven by yield trends, due to improved yield potential and crop management (Ewert et al., 2005) without being able to know if the trends are appropriate or consistent with the crop management evaluated (Shrestha et al., 2013). While our study did not consider nutrient limited growth, doing so would also represent a methodological challenge (Ewert et al., 2015). Consider that crop yield in crop models can be highly responsive to fertilizer application and this may interact under some conditions with climate and environment (Webber et al., 2015). However, in the subsequent economic optimization of yields in CAPRI, new fertilizer rates will be calculated and the resulting yields passed to the environmental model raising questions of whether iterations with the crop model are needed. Such questions deserve further research. Finally, our study is conducted at the regional scale and no attempt was made to quantify the uncertainty associated with aggregation of the climate and soil data (Zhao et al., 2015a; Hoffmann et al., 2016) or assumed crop

Table 9
Percent point differences (averages across crops/products) between actual (Act) and optimized (Opt) management cases of selected impact variables for different scenarios.

Impact variable	Scenario		
	A1B	B1	B2
	Act-Opt	Act-Opt	Act-Opt
Crop yield (CC, CO2)	− 15	− 14	− 15
Crop yield (CC, CO2, techn. trend)	− 15	− 14	− 15
Crop yield (CC, CO2, techn. trend, economic optimization)	− 4	− 7	− 7
Land use	0	2	2
Production	− 3	− 4	− 4
NH3 emission	0	− 3	− 2
N2O emission	− 1	− 1	− 2
N leaching + runoff	0	− 1	− 1

management (Patt et al., 2010). Likewise, the costs associated with adaptations such as changes in variety and sowing dates have not been explicitly considered (McCarl, 2007; Patt et al., 2010; Hertel and Lobell, 2014).

4.2. Magnitude of crop management impacts depends on the impact variable

The analysis reveals that the magnitude of impacts arising from the various crop management cases depends on the impact variable, and respective disciplinary model, being evaluated. Assumptions about crop management in terms of sowing date and thermal time requirements had large effects on the simulated yield changes due to climate change and CO_2, where Act yield changes were approximately 15% points smaller than Opt yield changes across crops and scenarios (Table 9). The systematically lower Act yield changes were partly compensated for through economic optimization in the economic model (average Act yield changes were lower by − 4 to –7% points after economic optimization). Additionally, land use partly increased to compensate for the still lower Act yields (land use change increased by up to + 2% points compared to the Opt management case), which led to changes in overall production that were smaller by − 3 to − 4% points on average (Act compared to Opt). The weaker response of the economic variables to the crop management variants propagated to the environmental impacts that varied between 0 to − 3% points difference between Act and Opt assumptions. While similar systematic differences between assumptions about crop management have been reported in other crop modelling studies (Teixeira et al., 2017; Webber et al., 2016a), this study additionally used a suite of different disciplinary models in linear succession, which influenced the diminishing importance of considering crop management specifications moving from crop model impact variables, to the economic variables to finally the environmental variables (Nelson et al., 2013).

In the crop model, EU-wide yield changes (averaged across crops and scenarios) were approximately 5% points greater if likely adaptations in sowing date and varieties were considered compared to climate change impacts alone. The error in not accounting for crop management adaptation increased to 20% points across crops and scenarios if the Opt adaptation scenario was considered. The interpretation of these results is not straightforward, as they apply to water-limited yields which assume no other limiting factors, and the necessary changes in farm management might not be applicable in practice (Webber et al., 2014). Earlier sowing dates, for example, might not be realized, as soil conditions prevent the necessary field operations. Equally, a change in simulated water-limited yields, as presented in this study, is unlikely to equal the change in actual farmers' yields as various biophysical, economic, regulatory and cultural factors influence real crop management and the assumption that factors currently constraining farmers will also apply in the future may not be valid (Finger, 2010; Olesen et al., 2011). Accordingly, the relative yield changes under climate change may be expected to lie in between the values for Opt and Act management, representing optimistic and pessimistic bounding cases, respectively.

The changes in crop yields led to interrelated simultaneous changes in cropping areas and production in CAPRI. Importantly, a positive impact of climate change on crop yields at larger scales will, all other things equal, decrease market prices and thus counteract to some extent the incentives that the yield increase provides to farmers and vice versa (Schneider et al., 2011). As a result, yield changes were predicted to be lower than after the biophysical optimization only. Endogenous yield changes in the economic model tended to be lower under Act than under Opt management. The lower yields under Act management were partly compensated for by cultivating more land of the respective crops. In general, the impacts of different global population and GDP assumptions on the economic indicators appeared to be much stronger than the impacts of the crop management specifications. This confirms the relative importance of the socio-economic environment in which climate change occurs compared to the biophysical impacts alone (Schmidhuber and Tubiello, 2007; Shrestha et al., 2013; Nelson et al., 2013).

In the environmental analysis, higher crop yields and corresponding higher nutrient demands led to strong increases in N fertilizer input. Under Opt management, this effect was stronger than under Act management in view of the higher yield levels.

4.3. Relative importance of technology progress versus climate change and management specifications

The estimated relative slopes for major crops indicate modest yield changes over the last two decades (Ewert et al., 2005), which contrast with higher growth in 1970s and 1980s. Evans (1997), as well as Finger (2010) in Switzerland, relate the declining trends to changes in agricultural policies for environmental protection, among other factors. Still, even these modest trends found in the last two decades lead to a potential large yield increase when accumulated until the year 2050. Crop models generally do not model the underlying changes in farm management (e.g. increased use of precision farming, improvements of cultivars, better weed and pest control). Rather, following Ewert et al. (2005), the historic trends were propagated into the future according to the assumptions in the respective scenario related to macroeconomic developments.

Assuming modest differences in yield growth depending on the macroeconomic development led to considerable differences in relative yield changes that were larger than the impacts of climate change or crop management specifications for all crops and scenarios considered. Our results suggest that in the longer term, technology progress mediated through GDP growth would have a larger impact on crops yields than climate change, even when Opt crop management is adopted. There is a scientific basis to expect continued yield growth due to progress in crop breeding. In wheat, for example, using more of the natural variability in Rubisco catalytic performance (Parry et al., 2007) combined with improved sink capacity by lengthening the duration of stem elongation (Slafer et al., 2001) and limiting lodging are all foreseen to improve yield potential on a time frame of 5–25 years (Reynolds et al., 2009). Despite this scientific potential, there is great uncertainty in knowing how economic and evolving crop preference factors will influence breeding progress in specific crops with other datasets already pointing to stagnation of yield trends in many parts of the world such as with wheat yields in Europe (Ray et al., 2012).

5. Conclusions

Our analysis clearly shows that crop sowing dates and thermal time requirements affect crop yields, land use, production and the environment. If sowing dates and crop cultivars were selected to optimize yields, yield changes were most positive for almost all crops and scenarios as compared to other management assumptions, but the effects were considerably smaller than assumptions about technological change. The economic model simulations showed that even after additional endogenous optimization, the differences between Opt and Act yields persisted. The impacts of different global population and GDP scenarios on the economic indicators appeared to be stronger than the difference in impacts of different management assumptions. The relative changes in N inputs by manure and NH_3 emissions were mainly determined by predicted changes in meat and milk production by the economic model. In contrast, the relative changes in N2O emissions and N leaching plus runoff were mainly determined by changes in total N inputs, being the sum of N fertilizer and N manure inputs, where N fertilizer inputs reflect predicted changes in crop yields and cropping patterns by the crop and economic model. Importantly, effects of management assumptions were most pronounced for crop yields and less for economic and environmental variables pointing to physical and economic adjustments along the model chain which are not yet well understood and modelled and which deserve attention in future research.

Overall, our paper underlines that an integrated assessment of future changes in crop and farm management, such as crop sowing dates and thermal time requirements investigated here, is necessary for a more comprehensive analysis of climate change impacts on yields, land use, production and environment. Such assessments for Europe should also account for interactions between economic growth, technological development and climate change. As technological development appears to be a key determinant of future agriculture, more research on what drives long-term technological progress in agriculture and yield gaps is necessary.

More broadly, there is a number of unresolved conceptual, methodological and technical issues that constrain integrated assessment and modelling activities. In the present study, models were loosely coupled based on output-input variable passing and feedbacks between models were not accounted for. A sound underlying methodology for such complex model linkages is not available yet but urgently needed to advance modelling of integrated problems in agriculture.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.agsy.2017.07.007.

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